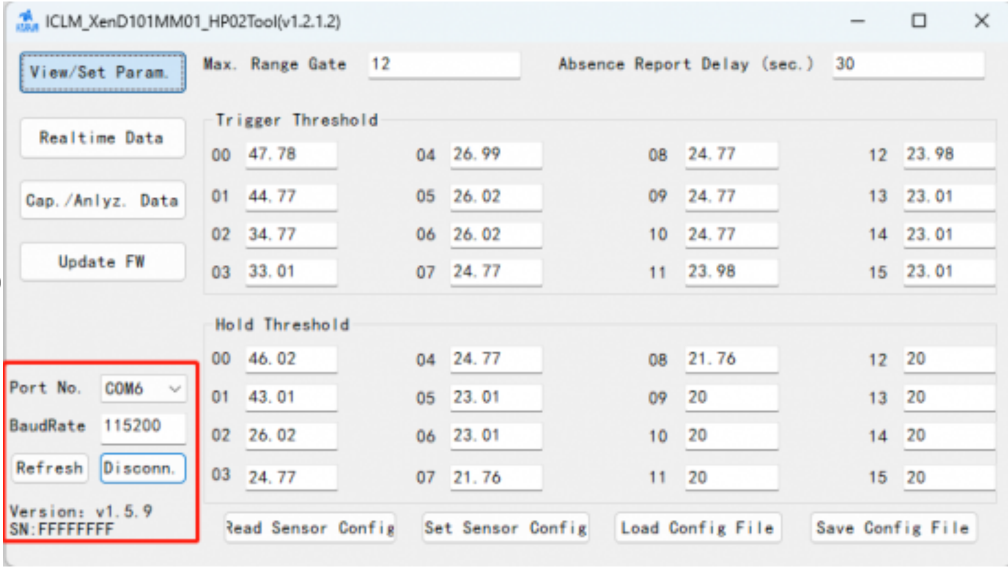
mmWave-hptool.png)



(/wiki/File:HMMD-mmWave-hptool01.png)

As shown in the above image, the host tool is divided into three areas: Device Operation Area 1, Function Button Area 2, and Function Page Area 3.

After connecting the host and the module, In Area 1 of the interface, the firmware version and serial number of the radar module will be displayed (if the serial number is not flashed, the host computer will display FFFFFFFF). The Function Page Area of the "Parameter View/Set" function will show the current settings of the radar module.

Terminology Explanation

- Maximum Distance Gate: Used to set the module's farthest effective detection distance, with one gate length equal to 70cm. Parameter range: 0~15.
- Target Disappearance Delay Time (s): The time it takes for the target state to switch from presence to absence. During this delay, if a person is detected, the timer resetting the delay time is triggered. After the module detects the absence state for a duration equal to the delay time, it reports no presence, and at this point, the OT2 pin outputs a low level. Parameter range: 0~65535.

Communication Protocol

The default baud rate of the radar serial port is 115200, 1 stop bit, no parity bit.

Protocol Format

HMMD mmWave Sensor data communication uses the small end format and all data is in hexadecimal.

Send Command with ACK

1. Read Firmware Version Command

This command reads the radar firmware version information.

Command Code: 0x0000

Command Value: None

Return Value: Version Number Length (2 bytes) + Version Number Byte String

Sending data:

FH(frame header) Byte 1~4	In-frame Data Length Byte 5,6	In-frame Data Byte 7,8	Frame Tail Byte 9~12
FD FC FB FA	02 00	00 00	04 03 02 01

Module ACK: (Successful Example)

Byte 1~4	Byte 5,6	Byte 7,8	Byte 9,10	Byte 11,12	Byte 13~18	Byte 19~24
FD FC FB FA	0C 00	00 01	00 00	06 00	76 31 2E 35 2E 35	04 03 02 01

2. Read Serial Number Command

This command reads the serial number of the radar.

Command Code: 0x0011

Command Value: None

Return Value: 2 bytes ACK status (0 for success, 1 for failure)

Sending Data:

Byte 1~4	Byte5,6	Byte 7,8	Byte 9~12
FD FC FB FA	02 00	11 00	04 03 02 01

Module ACK: (Successful Example)

Byte 1~4	Byte 5,6	Byte 7,8	Byte 9,10	Byte 11,12	Byte 13~14	Byte 15~18
FD FC FB FA	08 00	11 01	00 00	02 00	CB AB	04 03 02 01

3. Read Register Command

This command reads the registers of the radar.

Command Code: 0x0002

Command Value: 2 bytes chip address + (2 bytes address) * N

Return Value: (2 bytes data) * N

Sending Data:

Byte 1~4	Byte 5,6	Byte 7,8	Byte 9~12	Byte 13~16
FD FC FB FA	06 00	02 00	40 00 40 00	04 03 02 01

Module ACK: (Successful Example)

Byte 1~4	Byte 5,6	Byte 7,8	Byte 9,10	Byte 11,12	Byte 13~16

FD FC FB FA	06 00	02 01	00 00	07 02	04 03 02 01

4. Read Parameters Configuration Command

This command reads the configuration parameters of the radar.

Command Code: 0x0008

Command value: (2 byte parameter ID)*N

Return value: (4 byte parameter value)*N

Sending Data:

Byte 1~4	Byte 5,6	Byte 7,8	Byte 9~10	Byte 11~14
FD FC FB FA	04 00	08 00	01 00	04 03 02 01

Module ACK: (Success, maximum distance gate set to 12)

Byte 1~4	Byte 5,6	Byte 7,8	Byte 9,10	Byte 11,14	Byte 15~18
FD FC FB FA	08 00	08 01	00 00	0C 00 00 00	04 03 02 01

Module Report Data

The HMMD mmWave Sensor operates in default mode by outputting detection results through the serial port. The output consists of a string indicating ON/OFF along with the target distance gate.

In typical application scenarios, the host computer obtains data during the module's processing. Therefore, the module offers debug mode, report mode, and normal mode.

Normal Mode

Byte 1~4	Byte5,6	Byte 7,8	Byte9~10	Byte 11~14	Byte 15~18
FD FC FB FA	08 00	12 00	00 00	64 00 00 00	04 03 02 01

Debug Mode

Byte 1~4	Byte5,6	Byte 7,8	Byte9~10	Byte 11~14	Byte 15~18
FD FC FB FA	08 00	12 00	00 00	00 00 00 00	04 03 02 01

Debug Mode serial port data frame example：

PS: In debugging mode, the RDMAP data of all the range gates of a certain chirp are sent out each time, i.e., the data of all the 16 range gates of the first chirp will be sent out first, then goes the second chirp, and so on.

Header	Intra-frame Data	Tailer
AA BF 10 14	RDMAP: 20(Dopple)*16（number of range gate）*4（square of the amplitude)	FD FC FB FA

Report Mode

Byte 1~4	Byte5,6	Byte 7,8	Byte9~10	Byte 11~14	Byte 15~18
FD FC FB FA	08 00	12 00	00 00	04 00 00 00	04 03 02 01

In report mode serial port data frame example:

Frame Header	Length	Detection Result	Target Distance	The energy values for each distance gate	Frame Tail
F4 F3 F2 F1	2 bytes, total number of bytes for detection result, target distance, and energy values for each distance gate	1 byte, 00 absent, 01 present	2 bytes indicating the distance of the target phase from the radar in the scene	32 bytes, 16 (total number of distance gates) * 2 bytes, size of energy value for each distance gate from 0 to 15	F8 F7 F6 F5

Detection Range

The HMMD mmWave Sensor supports both top and wall mounting and is recommended for top mounting.

Top-mount Installation

The recommended top-mounted installation height is 2.7~3.0m. The top-mounted HMMD mmWave Sensor radar module, in the default configuration, has a maximum motion detection range within a conical three-dimensional space with a bottom radius of 5m.